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ORIGINAL ARTICLE

Patterns of Gene Expression in a Scleractinian Coral Undergoing Natural Bleaching

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Abstract Coral bleaching is a major threat to coral reefs worldwide and is predicted to intensify with increasing global temperature. This study represents the first investigation of gene expression in an Indo-Pacific coral species undergoing natural bleaching which involved the loss of algal symbionts. Quantitative real-time polymerase chain reaction experiments were conducted to select and evaluate coral internal control genes (ICGs), and to investigate selected coral genes of interest (GOIs) for changes in gene expression in nine colonies of the scleractinian coral *Acropora millepora* undergoing bleaching at Magnetic Island, Great Barrier Reef, Australia. Among the six ICGs tested, glyceraldehyde 3-phosphate dehydrogenase and the ribosomal protein genes S7 and L9 exhibited the most constant expression levels between samples from healthy-looking colonies and samples from the same colonies when severely bleached a year later. These ICGs were therefore utilised for normalisation of expression data for seven selected GOIs. Of the seven GOIs, homologues of catalase, C-type lectin and chromoprotein genes were significantly up-regulated as a result of bleaching by factors of 1.81, 1.46 and 1.61 (linear mixed models analysis of variance, $P<0.05$), respectively. We present these genes as potential coral bleaching response genes. In contrast, three genes, including one putative ICG, showed highly variable levels of expression between coral colonies. Potential variation in microhabitat, gene function unrelated to the stress response and individualised stress responses may influence such differences between colonies and need to be better understood when designing and interpreting future studies of gene expression in natural coral populations.

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Introduction

The future of coral reefs is uncertain because many reef-building corals live close to their upper thermal limit (Jokiel and Coles 1990; Berkelmans and Willis 1999; Podesta and Glynn 2001) and may be unable to adapt to the rapidly warming seas. Global surface temperature (over land and sea) has increased ~0.2°C per decade over the past 30 years (Hansen et al. 2006) and is projected to increase by another 0.4°C over the next 20 years (Bates et al. 2008). These environmental changes will most likely translate into substantial loss of coral reef biomass and diversity (Connell and Slatyer 1977; Maynard et al. 2008). To date, loss of coral cover has mainly occurred through mass coral bleaching events that will likely increase in intensity and frequency with increasing sea surface temperatures (reviewed in Brown 1997; Hoegh-Guldberg 1999).

Coral bleaching is the breakdown of the symbiosis between a coral host and its endosymbiotic dinoflagellates (*Symbiodinium* spp., aka zooxanthellae), which results in coral depigmentation via the loss of *Symbiodinium* cells and/or their pigments. As reef-building corals receive most of their nutrition from *Symbiodinium* photosynthesis (Muscatine and Porter 1977), the loss of symbiont cells deprives the coral of essential nutrients leading to reduced growth, decreased reproductive output and/or mortality (reviewed in Glynn 1993). The available evidence suggests that one key trigger for coral bleaching is damage to the photosynthetic apparatus (and/or downstream enzymes) of the dinoflagellate symbionts and subsequent cellular damage resulting from continued absorption of excitation (light) energy beyond the processing ability of heat affected symbionts (Iglesias-Prieto et al. 1992; Lesser 1996, 1997; Jones et al. 1998; Warner et al. 1999). The cellular damage triggering bleaching has been widely attributed to increased production of reactive oxygen species (ROS) and the inability of detoxifying antioxidant enzymes to keep pace with the rate at which ROS are produced (Lesser et al. 1990; Lesser 1997; Nii and Muscatine 1997; Downs et al. 2002; Dunn et al. 2002; Franklin et al. 2004; Smith et al. 2005).

Endogenous antioxidant enzymes, including superoxide dismutases and catalase, are produced as part of the first line of defence against excessive ROS production in symbiotic cnidarians (Dykens and Shick 1982; Nii and Muscatine 1997; Brown et al. 2002; Richier et al. 2003; Merle et al. 2007). This makes these enzymes potentially sensitive indicators for the onset of bleaching stress in reef-building corals. The role of heat, light and ROS in triggering the bleaching response of corals is increasingly evident but many novel stress response pathways remain to be discovered and understood. For instance, recent studies raise interesting questions about the potential for green fluorescent protein (GFP) homologues to protect corals against heat-induced bleaching. Scleractinian corals have a diverse complement of GFP-like proteins and non-fluorescent chromoprotein homologues (Dove et al. 1995; Mazel 1995; Dove et al. 2001; Miyawaki 2002; Mazel et al. 2003). These are located within the tissue of the host (Mazel 1995; Salih et al. 2000) and their localisation to vesicles above *Symbiodinium* cells in high-light environments is suggestive of a photoprotective role for the symbionts (Salih et al. 2000). A GFP from the hydromedusa, *Aequorea victoria*, quenches superoxide radicals in vitro indicating that some of these proteins could also act in an antioxidant capacity (Bou-Abdallah et al. 2006). The putative photoprotective role of GFP-like proteins, and their ability to confer any protection against bleaching, remain however both uncertain and contentious (Mazel et al. 2003; Dove 2004; Dove et al. 2006; Smith-Keune and Dove 2008).

General stress response proteins such as heat shock proteins and antioxidant enzymes have both been identified in bleaching corals in the past, and quantification of these proteins has been proposed with a view to using protein-based assays as sensitive biomarkers for diagnosing or identifying stress (reviewed in van Oppen and Gates 2006). However, changes in gene transcript levels usually precede those in protein levels; therefore, changes in gene expression may serve as even more sensitive indicators of the onset of a bleaching stress. Using this approach to understand coral stress responses is only recently becoming possible with the availability of large databases of expressed sequence data for certain scleractinian corals (Edge et al. 2005; Technau et al. 2005; Meyer et al. 2009) and other symbiotic cnidarians (Sunagawa et al. 2009). With this in mind, *Acropora millepora* represents an ideal model species, as it is easy to identify, occurs with high abundance on the Great Barrier Reef (GBR) and throughout the Indo-West Pacific, is common on inshore reefs of the GBR and is the most represented species in sequence databases. Inshore reefs are generally the most impacted by anthropogenic stressors and consequently are the most extensively and severely impacted by recent mass bleaching events (Berkelmans and Willis 1999; Berkelmans et al. 2004).

The identification of appropriate internal control genes (ICGs), for which expression patterns remain stable despite the physiological stress and substantial loss of algal symbionts associated with bleaching, is essential if gene expression assays are to be used as sensitive tools for coral bleaching field studies. The present study is the first to develop and validate quantitative polymerase chain reaction (qPCR) assays for biomonitoring of genes of interest (GOI) expression levels in a widespread, Indo-Pacific scleractinian coral (*A. millepora*) during a natural bleaching event. We identify three suitable ICGs for this species as well as two novel candidate stress response genes for future study.

Materials and Methods

Sample Collection Thirty *A. millepora* colonies were tagged between 1.5 and 3 m depth on the reef flat of Nelly Bay, Magnetic Island (19°10′S, 146°50′E), in the central GBR, Australia. Samples were collected at 21 time points over a 2-year period. Sampling started in November 2000 and covered a bleaching event from January to March 2002. Small branch fragments, ~5 cm long, were obtained from the centre of tagged colonies at approximately mid-day on each sampling occasion. Branches were promptly snap-frozen in liquid nitrogen at the end of each dive and brought back to the laboratory to be stored in a −80°C freezer until molecular analysis. Samples from a subset of nine colonies, from two time points, the 24th of January 2001 and the 24th of January 2002, were used for an assessment of gene expression levels in this study. These represent a non-bleaching and bleaching summer, respectively.

Water temperatures were obtained from the Great Barrier Reef Marine Park Authority (GBRMPA) water temperature monitoring programme. The GBRMPA temperature logger on the Nelly Bay reef flat was located ~50–100 m from the tagged colonies. Daily mean, minimum and maximum temperatures on the Nelly Bay reef flat were calculated for the duration of the monitoring period from half-hourly logger data.

Bleaching Condition: Symbiodinium Cell Counts and Chlorophyll Content The extent and severity of coral bleaching was monitored for each colony by measuring algal symbiont density and the ratio of visually degenerate to healthy symbionts as described in Bourne et al. (2008). Chlorophyll *a* content of algal cells was also measured in a sub-sample of isolated *Symbiodinium* for each colony using methanol extraction. Sub-samples were centrifuged at 250×*g* for 10 min, and algal cell pellets were resuspended in 4 ml of 100% methanol and extracted overnight at −20°C in the dark. Extracts were clarified by centrifugation at 500×*g* (10 min), and absorbance at 635 and 668 nm was measured in a quartz cuvette of 1-cm path length. Chlorophyll *a* content was calculated using the equation: $13.8\times A_{668}-1.3\times A_{635}$ (Jeffrey and Haxo 1968). For each sub-sample, the values from two consecutive extracts of algal cell pellets were combined, adjusted for total sample volume, and the data were normalised to estimated numbers of algal cells (both healthy and degenerate).

Total RNA Extraction Total RNA was extracted using TRI reagent (Ambion, Australia) in an optimised protocol for adult coral tissue as follows. Frozen pieces of coral branch were first crushed in liquid nitrogen (LN_2) in a pre-chilled stainless steel mortar and pestle. Samples were crushed with a bench press (LabTek) that applied 12 tonnes of pressure reducing the samples to a 0.5-mm chip, subsequently loosened to a fine powder using a ceramic pestle. Two millilitres of TRI reagent were added to ~100 mg of powder per sample and vortexed at low speed for 5 min at room temperature. Samples were spun at 12,000×*g* for 10 min at 4°C to pellet skeletal material. The TRI reagent manufacturer's protocol was then followed to isolate RNA from the aqueous phase with the inclusion of an additional chloroform separation step to reduce mucosal polysaccharide contamination. RNA samples were DNase-treated using the DNA-free™ kit (Ambion). Integrity of the total RNA was determined by running RNA 6000 nano assays on the Bioanalyzer (Agilent Technologies). Integrity values over 6.0 reflect minimal degradation of the RNA and are suitable for downstream gene expression applications. For samples used in this study, integrity values ranged between 6.1 and 9.5. Nucleic acid concentrations were measured at 260 nm with the ND-1000 Spectrophotometer (NanoDrop Technologies). Total RNA samples with a 260 nm/280 nm ratio falling between 1.8 and 2 were considered pure and included the nine selected colonies.

Selection of ICGs and GOIs qPCR is complicated in symbiotic organisms because of the potential contribution of each partner to the total pool of nucleic acid (Mayfield et al. 2009). Therefore, in addition to being stable under the experimental conditions, the primers used to amplify ICGs must be specific to the host. To meet these criteria, ICG expression stability was determined by pairwise comparison of expression levels between all candidate reference genes, using the same amount of total RNA in each qPCR reaction for bleached and unbleached samples of the same colony. Furthermore, host-specific primers were designed in open-reading frames of aposymbiotic (devoid of symbionts) larval *A. millepora* expressed sequence tag (EST) sequences (Grasso et al. 2008) and checked for specificity in silico using the Primer-BLAST tool from NCBI. Extensive mechanical grinding was avoided to limit breakage of the symbiont cell wall and minimise contamination from *Symbiodinium* RNA. Symbiont contamination was evaluated with *Symbiodinium*-specific primers (ESM Table 1) on three pooled cDNAs from aposymbiotic planula larvae, juvenile and adult tissues, harbouring low and high *Symbiodinium* densities, respectively. Despite efforts to minimise contamination, non-normalised values for gene expression of host genes were likely to be somewhat overestimated due to a higher host/symbiont RNA ratio in bleached versus non-bleached samples, but this error appears to be minimal (ESM Fig. 1). Following DNase treatment, no genomic DNA contamination could be detected from either the host or the symbionts using intron-annealing primers in standard PCR reactions.

Beta actin, glyceraldehyde 3-phosphate dehydrogenase (GAPDH) and the ribosomal protein L13a (rpL13a) are commonly used ICGs in gene expression experiments (Vandesompele et al. 2002) and were therefore considered in this study. Other potential ICGs were also evaluated including a coral homologue of adenosylhomocysteinase (Levy et al. 2007), filamin, tropomyosin, rpL9, rpS7 and two unknown sequences, Ctg1913 and Ctg3235. These genes were selected based on their stable expression at the time points used in this study in a small pilot microarray experiment (data not shown) based on a method of reference gene selection similar to that of Rodriguez-Lanetty et al. (2008). A quick test using a subset of samples in preliminary qPCR runs isolated the following candidate ICGs: GAPDH, rpL13a, rpL9, rpS7, Ctg1913 and Ctg3235, which were thoroughly tested in the full-scale qPCR experiment.

GOIs were selected based on their known or suspected function relevant to the coral bleaching response (e.g. Morgan et al. 2005; Desalvo et al. 2008; Edge et al. 2008). More specifically, we selected homologues of catalase (*Am*Cat), ferritin (*Am*Ferri) and selenoprotein W (*Am*SW) as indicators of oxidative stress (reviewed in Whanger 2000; Lesser 2006; Mladenka et al. 2006). Further, the chromoprotein (*Am*Ch) represents a host pigment and adds information to the work performed on GFP-like genes in corals (Dove 2004; Bou-Abdallah et al. 2006). C-type lectin (*Am*CTL), tetraspanin CD151 (*Am*CD151) and tribble (*Am*Trib) were chosen as potential indicators of stress or an immune response (Kvennefors et al. 2008), cell–cell adhesion (Shigeta et al. 2003) and mitogen-activated protein kinase (MAPK) regulation (Kiss-Toth et al. 2004), respectively. Defence mechanisms may play an important part in the bleaching stress response, as bleached corals have been found to be more susceptible to disease (Muller et al. 2008). One hypothesis for this increased susceptibility is an increased microbial community due to elevated temperature combined with a decrease in coral tissue thickness due to bleaching, leading to bacterial infection. Therefore, regulation of genes with functions in immunity, such as C-type lectins, is expected. In addition, symbiont-containing cell detachment has been observed as one of the cellular mechanisms involved in coral bleaching in the field (Brown et al. 1995). Therefore, cell–cell adhesion-related genes, such as tetraspanins, represent good targets. Finally, genes regulating MAPK expression, such as tribble, are also of interest because MAPKs are present in corals and may play a role in osmoregulatory cascades during coral bleaching (reviewed in Mayfield and Gates 2007).

Two-step qPCR In order to validate the stability of six candidate ICGs and the change in expression for seven GOIs, qPCR assays were performed on nine individual colonies. The same amount of pure total RNA per colony was reverse-transcribed to cDNA in a Mastercycler personal (Eppendorf, Hamburg, Germany) following the SYBR GreenER™ Two-Step qRT-PCR universal kit protocol (Invitrogen, Australia). cDNA samples were diluted five-fold prior to qPCR runs. Primer pairs were designed (see ESM Table 1 for primer sequences) to amplify fragments of between 90 and 110 bp, with a GC content ranging from 50% to 60% and a melting temperature between 57 and 63°C using the OligoPerfect™ designer software (Invitrogen). Primers were synthesised commercially (GeneWorks Pty Ltd, Hindmarsh, Australia). An optimum annealing temperature of 60°C was determined on the Mastercycler Gradient (Eppendorf). After UV sterilisation of all blocks and tubes, qPCR reactions were loaded in triplicate using the CAS-1200 loading robot (Corbett Research) to minimise pipetting error. Assays were performed following the programme proposed in the SYBR GreenER™ Two-Step qRT-PCR kit (Invitrogen) on the Rotor-Gene 3000 (Corbett Research). The master-mix was scaled down to 20 µl as follows: 10 µl SYBR GreenER™ qPCR SuperMix (Invitrogen), 1 µl of each primer (at 4 µM), 2 µl of 5× diluted cDNA and 6 µl diethylpyrocarbonate-treated water. Runs were followed by melting curves for detection of any non-specific amplification or primer dimerisation, with heating starting at 57°C and a rate of 0.1°C per second up to 95°C with continuous measurement of fluorescence. The efficiency of each primer was evaluated using the dilution method (Rebrikov and Trofimov 2006) on a pool of all cDNA samples and used in the calculation of gene expression level using the normalised relative quantification method as in Hellemans et al. (2007).

Data on the expression levels of each sample were obtained in the form of quantification cycle (Cq) by manually positioning the lowest possible threshold above background fluorescence on the exponential phase of all amplification plots viewed using the logarithmic scale for the fluorescence axis. This provides the most accurate Cq measurements as this part of the plots represents the optimum qPCR kinetics (Bustin 2004).

Statistical Analysis Hellemans et al. (2007) suggested two measures to assess a reference gene for qPCR: (1) the coefficient of variation (CV) of the normalised relative quantities in all conditions and (2) a stability value, *M*, that indicates the stability of the non-normalised relative quantities of a transcript with respect to the other transcripts. In their experiment with heterogeneous samples, Hellemans et al. (2007) observed that *M* and CV could be values up to 1 and 50%, respectively, and advised the use of at least three genes for the normalisation of qPCR data. The qPCR data were therefore transformed and normalised according to the methods proposed by Hellemans et al. (2007) using the ICGs described below, with *M* and CV values lower than 1 and 50%, respectively. Differences in gene expression were assessed with linear mixed models analysis of variance (Demidenko 2004) using condition (healthy-looking or severely bleached) as the fixed effect, colony as the random effect and the log of the transcript relative quantity as the dependent variable. The statistical analysis was conducted using the R statistical environment (R_Development_Core_Team 2008). Statistical significance was evaluated at the α=0.05 level for the *F* statistic.

Results

Changes in Symbiodinium Densities, Chlorophyll a Content and Water Temperature At the January 24th 2001 time point, the average symbiont density was 2.07×10^6 ($\pm0.25\times10^6$) cells cm^{-2} (Fig. 1a) and none of the nine colonies showed visual signs of bleaching (Smith 2005). Furthermore, on the preceding sampling date (3 November 2000), the same colonies looked healthy and the average symbiont density was 2.50×10^6 ($\pm0.37\times10^6$) cells cm^{-2}. In comparison to the overall fluctuation in symbiont population over the 2-year experimental period (Bourne et al. 2008), these values fall within the range of *Symbiodinium* densities found in healthy-looking individuals. An average symbiont density of 1.10×10^6 ($\pm0.18\times10^6$) cells cm^{-2} was measured in the same colonies exactly a year later, at the January 24th 2002 time point (Fig. 1a). This corresponds to a drastic drop, which was emphasised by severe loss of colour in all nine colonies. The symbiont density and chlorophyll *a* concentration were both significantly reduced in 2002 (paired two-tailed *t* test, $P<0.05$; Fig. 1). The water temperature monitoring data revealed that temperature stayed below 29°C for 3 weeks prior to sampling in 2001. However, during an equivalent period of time prior to the 2002 sample, water temperature rose to 32°C over a few days and reached a plateau for 2 weeks before dropping to 30°C during the third week. Bleaching was first observed on 7 January 2002, ~2 weeks prior to the collection of 24 January 2002 (Bourne et al. 2008).

ICGs, Differentially Expressed GOIs and Inter-colony Variability Consistent differences in transcript quantities between conditions (bleached and non-bleached) were observed among the six candidate ICGs for almost all colonies under investigation, indicating stability in expression (ESM Fig. 1). Moreover, GAPDH, rpS7 and rpL9 had the best CV (29–32%) and *M* (0.58–0.62) values (ESM Table 2), and were therefore used for normalisation of GOI-expression data (Fig. 2).

Using mixed model analysis of variance, three genes were found to be significantly differentially expressed in the qPCR experiment: *Am*Cat (F=19.8, $P=6.6\times10^{-5}$), *Am*CTL (F=13.8, $P=5.8\times10^{-4}$) and *Am*Ch (F=12.5, $P=9.7\times10^{-4}$). These three genes were up-regulated in bleached colonies by factors of 1.81, 1.46 and 1.61, respectively (Table 1), relative to the three best ICGs mentioned above.

A significant difference among colonies was detected for two putative GOIs (*Am*Trib, F=89.18, $P=8.0\times10^{-5}$; *Am*SW, F=6.9, $P=3.4\times10^{-2}$) and one potential ICG, Ctg1913 (F=15.6, $P=5.5\times10^{-3}$). This inter-colony variability is not only caused by differences in amplitude of the response to bleaching but also by dramatic changes in the direction of individual regulation (Fig. 3).

Discussion

Gene expression studies using qPCR can help identify the onset of the physiological changes indicative of stress that leads to the phenomenon of coral bleaching. However, ICGs that exhibit stable expression under field conditions are essential if gene expression is to be used as an indicator for the onset of stress. We have identified three such genes from a widely distributed Indo-Pacific coral, which are therefore promising candidates for the normalisation of gene expression data during natural coral bleaching events. Furthermore, our finding of three up-regulated GOIs (catalase, chromoprotein and C-type lectin homologues) in naturally bleached colonies of *A. millepora* supports and enhances our understanding of the cellular processes underpinning bleaching. Large inter-colony variation in the expression patterns for three other genes, *Am*Trib, *Am*SW and an unknown gene Ctg1913, were observed during this natural bleaching event, highlighting the need for appropriate gene selection and adequate sample sizes for future gene expression studies on natural coral populations.

ICGs for Coral Bleaching ICGs were selected based on their consistency in expression despite the drastic decrease in symbiont densities between bleached and non-bleached samples and technical discrepancies across samples using the gNorm method (Vandesompele et al. 2002). The

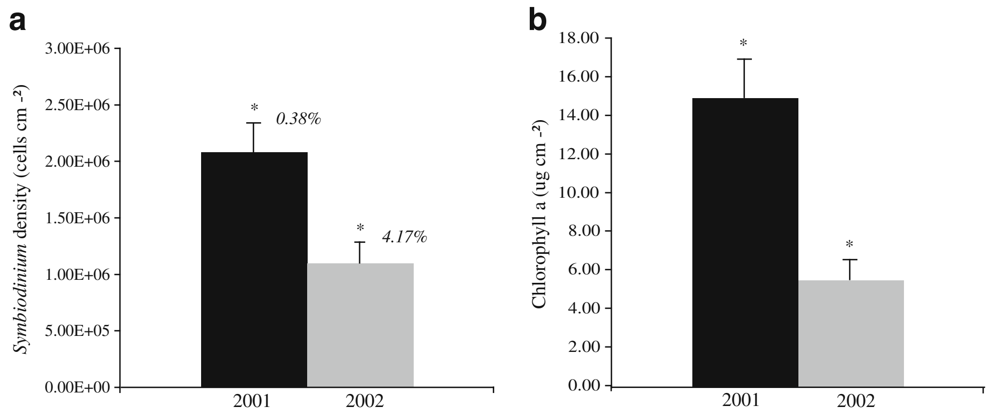

Fig. 1 Comparison of **a** the mean *Symbiodinium* cell densities and **b** chlorophyll *a* concentrations in samples used in the qPCR experiment (n=9) at two time points, 24 January 2001 (*2001*) and 24 January 2002 (*2002*). *P<0.05, paired two-tailed t test. The percentages of degenerate *Symbiodinium* cells are indicated

ribosomal protein rpL9 and rpL13a genes encode components of the 60S subunit and therefore function together, while the rpS7 protein is a component of the 40S subunit. Nevertheless, the expression of rpS7 and rpL9 appeared to be more stable than rpL13a and were therefore used in the present analysis. In support of this, another 60S subunit ribosomal protein gene, rpL12, has been suggested as a potential ICG in cnidarian studies (Rodriguez-Lanetty et al. 2008). GAPDH, also a commonly used ICG, was relatively stable in its expression between bleached and unbleached *A. millepora* colonies and was therefore included in the ICG panel for normalisation.

In model organisms, GAPDH, beta actin and the 16S ribosomal subunit are commonly used ICGs in qPCR gene expression assays (Vandesompele et al. 2002). Although these genes are suitable ICGs for some controlled laboratory experiments, they are not uniformly applicable across all species, individuals, tissues and experimental conditions. For example, beta actin has been used to normalise qPCR gene expression data from both *A. millepora* (Levy et al. 2007) and other cnidarians (Moya et al. 2008; Rodriguez-Lanetty et al. 2008); however, this gene did not perform well in preliminary trials of the field-bleached colonies sampled here (data not shown) and was found significantly regulated in a recent coral bleaching microarray study (Seneca et al., unpublished).

Additional challenges are present when identifying stably expressed ICGs for qPCR assays in field coral samples. For corals undergoing bleaching in the field, individuals from the same sampling site (tens of square metres) may experience strong microhabitat differences (e.g. light and temperature exposure), which may cause subsequent differences in their responses. In some species, multiple *Symbiodinium* types (with varying physiological performances) may co-exist on the same reef producing staggered phenotypic responses to widespread stressors (Rowan et al. 1997; Jones et al. 2008). Genetically determined variation in bleaching susceptibility of the coral host may further contribute to variance in the bleaching response among colonies. As a result, qPCR assays on coral field samples need to include multiple carefully selected ICGs.

In the present study, we demonstrated that our samples were contaminated with *Symbiodinium* cDNA. Primers for the proliferating cell nuclear antigen gene in *Symbiodinium* (Boldt et al. 2009) failed to amplify cDNA from aposymbiotic planula larvae (Cq>33) but amplified juvenile (Cq<23) and adult cDNAs (Cq<21). Naturally, one would expect this contamination to affect healthy and bleached tissues in such a way that the Cq values for host ICGs would be lower in bleached samples due to the dilution effect of higher symbiont density in healthy coral samples.

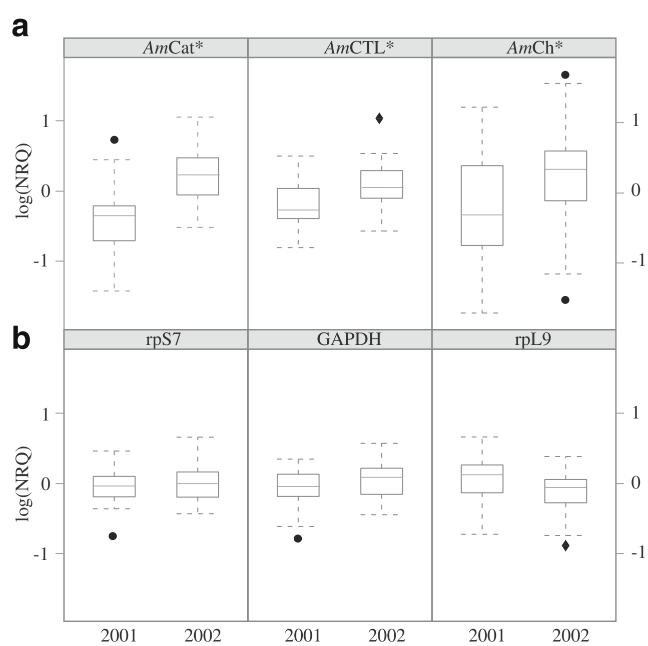

Fig. 2 Box plots comparing the log-normalised relative quantity (log (NRQ)) for nine *Acropora millepora* colonies in healthy-looking (*2001*) and severely bleached (*2002*) condition for: **a** three GOIs, catalase (*Am*Cat), C-type lectin (*Am*CTL) and chromoprotein (*Am*Ch), and **b** three ICGs, ribosomal protein subunit S7 and L9 (rpS7 and rpL9), and GAPDH. *P<0.05, linear mixed models analysis of variance (*black circles*, outliers; *black diamonds*, superposed outliers)

Table 1 Regulation of three GOIs (based on qPCR) in nine field colonies of *A. millepora* undergoing bleaching

Genes	GenBank accession number	*E* value	Fold change in bleached samples	*P* value
*Am*Cat	DY586920	$9\times e^{-70}$	1.81	$6.6\times e^{-5}$
*Am*CTL	DY587389	$6\times e^{-16}$	1.46	$5.8\times e^{-4}$
*Am*Ch	EZ032343	$3\times e^{-96}$	1.61	$9.7\times e^{-4}$
*Am*Trib	DY587605	$2\times e^{-7}$	na	na
*Am*CD151	DY586383	$2\times e^{-19}$	na	na
*Am*Ferri	DY586867	$1\times e^{-62}$	na	na
*Am*SW	DY585805	$1\times e^{-14}$	na	na

The same nine colonies were sampled exactly 1 year apart and were healthy looking at the first and severely bleached at the second sampling time point. The fold changes of gene expression are shown with the *P* values for those genes that were significantly differentially expressed. The *E* value corresponds to the best match of a Tblastx search in the nucleotide collection (nr/nt)

However, the small shift in Cq values for the best performing ICGs, GAPDH, rpL9 and rpS7 presented here was not uniform in direction across all colonies (ESM Fig. 1). This inconsistency in direction of the shift for Cq values between bleached and healthy samples suggests that the symbiont contamination is minimal and most likely confounded with other technical artefacts such as variations in cDNA efficiency and/or messenger RNA yield between samples.

Here, we present an alternative method of normalisation to that developed by Mayfield et al. (2009) which consists of using a spike of exogenous RNA/DNA to normalise for differential extraction of *Symbiodinium* nucleic acids in symbiotic systems. We suggest that one should develop an ICG system of at least three genes exhibiting stable expression levels throughout the experimental period before assessing normalised relative quantification of scleractinian coral gene expression. Potential ICGs can be selected using microarray data as described in Rodriguez-Lanetty et al. (2008) before being further tested and used in qPCR experiments, whether to validate microarray data as in Desalvo et al. (2008) or monitor gene regulation in the field as presented here. We do not claim that the ICGs presented here will be universally useful across conditions and species but, rather, that they are the best under the conditions in this study to accurately measure change in gene expression and should be considered as candidates in other studies, especially those using *A. millepora*.

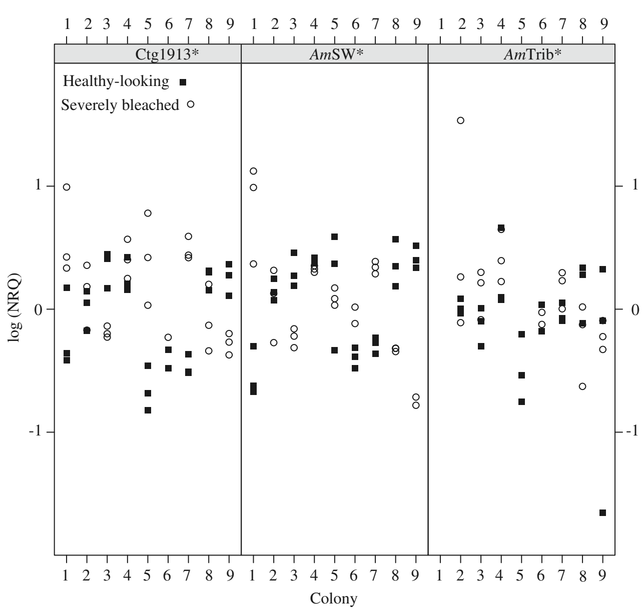

Fig. 3 Log (NRQ) data for three genes showing great inter-colony variability in expression among the nine colonies. *$P<0.05$, linear mixed models analysis of variance (qPCR reaction triplicates for one colony are depicted by the same symbol)

Regulation of GOIs during Bleaching The regulated genes discovered here are uncharacterised in corals, and little is known about their specific function in cnidarians. Their function in other organisms is summarised below. The consistent up-regulation of three genes in nine independent *A. millepora* colonies during a natural coral bleaching event suggests, however, that these three genes in particular warrant further investigation for their role in the bleaching process and their potential as useful stress indicators in field sampling studies.

AmCatalase

The most studied of our up-regulated genes is an *A. millepora* catalase homologue (*Am*Cat). Catalase is an antioxidant enzyme that limits the production of highly cytotoxic hydroxyl radicals (HO·) (Lesser 2006). Our finding of a significant up-regulation of catalase in *A. millepora* during natural coral bleaching supports previous studies indicating that oxidative stress likely affects host coral tissue during bleaching events (Lesser 1996, 1997; Lesser and Farrell 2004; Lesser 2006) and is consistent with its up-regulation in a time course experiment looking at heat-induced bleaching in laboratory conditions by Desalvo et al. (2008).

Catalase reduces the H_2O_2 produced in excess by photosynthetic organisms undergoing photoinhibition (reviewed in Smith et al. 2005) to water and molecular oxygen (Mladenka et al. 2006). During coral bleaching, heat and light may induce photoinhibition in endosymbiotic zooxanthellae, leading to the overproduction of ROS, which may diffuse through membranes and damage host cells, resulting in their detachment or in symbiont expulsion (Brown 1997; Fitt et al. 2001; Lesser and Farrell 2004; Tchernov et al. 2004). Although *Symbiodinium* possess antioxidant enzymes capable of detoxifying H_2O_2 (Matta and Trench 1991), overproduction of H_2O_2 may overwhelm the antioxidant defence of the symbiont and diffuse to the host through cell membranes (Lesser 2006). Therefore, significant up-regulation of catalase in the host tissue of field-bleached *A. millepora* supports a role for this antioxidant enzyme in the maintenance of homeostasis in response to oxidative stress during coral bleaching (Merle et al. 2007).

AmChromoprotein

We have also identified an up-regulated chromoprotein gene, *Am*Ch. Coral chromoproteins are close relatives of fluorescent proteins such as GFP, and they may play an important role in the stress response of corals. The up-regulated *Am*Ch gene examined here encodes a blue non-fluorescent chromoprotein (Beltran-Ramirez 2008) that was originally sequenced from *A. millepora* larvae (Technau et al. 2005) and which shares 95% amino acid homology (211/221 AA) with a blue chromoprotein (GenBank AY646075) absorbing at 588 nm first identified in adult *A. millepora* tissue (Alieva et al. 2008). Interestingly, a blue pigment appears more abundant in recently compromised tissue of *A. millepora* than in adjacent healthy tissue in the field (personal observation). Further work is needed to determine if this pigment corresponds to *Am*Ch. In addition to the substantial bleaching reported here, several of the *A. millepora* colonies included in the present study developed necrotic-looking lesions during the 2002 natural bleaching event (C. Smith-Keune, personal observation), and it is intriguing that this event also coincides with an increase in *Am*CTL gene transcript levels.

The traditionally proposed role of fluorescent and non-fluorescent chromoproteins in symbiotic corals was as a sunscreen or photoprotective shield for corals living in high-light environments (Kawaguti 1944; Salih et al. 2000; Dove et al. 2001). It has also been proposed that this role may be reversed to actually enhance light capture for corals in low-light conditions (Schlichter and Fricke 1990). However, it has been suggested that the putative photoprotective function of certain GFP-like proteins in acroporid corals may be rapidly lost during times of thermal stress (Dove 2004; Smith-Keune and Dove 2008), and several other studies argue against a direct role for coral GFP homologues in photoprotection or improving photosynthetic efficiency in coral symbiosis altogether (Gilmore et al. 2003; Mazel et al. 2003; Wiedenmann et al. 2004). On the other hand, a cyan fluorescent protein was differentially expressed in competent *Montastraea faveolata* larvae as a result of inoculation with symbionts (Voolstra et al. 2009). Interestingly, an antioxidant role via the quenching of superoxide radicals has recently been shown for a GFP from the hydromedusa, *A. victoria* (Bou-Abdallah et al. 2006). In light of this, the concomitant up-regulation of both the *Am*Ch and *Am*Cat gene transcripts during coral bleaching in the present study suggests that a similar antioxidant role for *Am*Ch in *A. millepora* could be considered.

At the same time, the up-regulation of gene expression for the blue chromoprotein gene *Am*Ch identified in field-bleached colonies here contrasts with other studies. Smith-Keune and Dove (2008) showed the rapid down-regulation of another GFP-like gene AmA1 (a homologue of the red fluorescent *Zoanthus* sp. protein zoan2RFP) in laboratory heat-stressed *A. millepora* colonies experiencing temperatures of 32 and 33°C. Similarly, a GFP-like homologue was down-regulated in *M. faveolata* colonies exposed to 32°C for 3 days in the laboratory (Desalvo et al. 2008). This may suggest a variable role for these GFP-like proteins within *A. millepora* host tissues or may indicate a difference in the bleaching mechanisms occurring in the field compared to that induced at similar temperatures in the laboratory for this species. For example, huge difference in light intensity between laboratory and field conditions may explain such opposite patterns in gene expression.

AmC-type Lectin

The qPCR assays showed consistent up-regulation of a putative C-type lectin gene (*Am*CTL). The *Am*CTL predicted peptide has a putative C-type lectin-like domain of the type found in lectins of other marine invertebrates, for example, CEL-1 from *Cucumaria echinata* and echinoidin from *Anthocidaris crassispina* (E value=$2e^{-20}$). In other animals, C-type lectins are involved in processes including extracellular matrix organisation, pathogen recognition and cell–cell interactions, which are all processes that may be relevant during coral bleaching events. At least seven other lectin-related genes are expressed in *A. millepora* including several that are expressed at the time of metamorphosis (Grasso et al. 2008). This suggests potential roles in cell recognition and lysis or in *Symbiodinium* recognition and incorporation. The closest known protein-encoding gene to *Am*CTL was a C-type lectin homologue (FJ628422, E value=$1e^{-37}$), which was found down-regulated prior to

Symbiodinium expulsion in *Pocillopora damicornis* colonies exposed to 32°C in the laboratory (Vidal-Dupiol et al. 2009). In addition, a mannose-binding lectin paralogue (GenBank ACF08844, *E* value=$2e^{-24}$) named millectin is 41% identical and 60% similar to *Am*CTL predicted peptide sequence. More importantly, both share conserved amino acids in the mannose- and the calcium-binding sites of C-type lectins involved in immunity (Kvennefors et al. 2008).

Inter-colony Variability in Levels of Gene Expression in a Natural Population A surprising degree of inter-colony variability in direction and magnitude of gene expression changes was observed for two of the GOIs, *Am*SW and *Am*Trib, and one candidate ICG, a coral-specific gene of unknown function, Ctg1913. Unfortunately, little is known of the function of any of these genes in other animals, and the homology of these genes to known orthologues in GenBank was generally low. Regardless of their functions, the highly variable expression of these genes in naturally bleached colonies of *A. millepora*, on the Nelly Bay reef flat during the bleaching event of 2002, suggests either substantial variation in microenvironment of the sampled colonies or different genetically determined individual colony responses. It is essential for valid biomonitoring studies that the chosen stress response genes show similar gene expression profiles for all or at least the majority of individuals on the reef under study. Consistent sampling of colonies with respect to overall morphology and replication at both inter- and intra-colony levels should be carefully considered in future field studies applying assays similar to those developed here.

We utilised a single branch from a similar location at the centre of each tagged *A. millepora* colony, which was sampled at approximately mid-day on each sampling date. The future inclusion of intra-colony replication, and measurements of protein levels in addition to gene expression levels, would substantially enhance our understanding of the biological significance of the changes in expression of the genes measured here. Recent gene expression studies on the same species have observed high levels of inter-colony variability (Bay et al. 2009a, b; Császár et al. 2009), making measurements of significant change in expression at the population level difficult, and indicating high biological replication is required. Despite the presence of similar levels of inter-colony variability observed here, we were able to successfully detect significant and consistent change in expression levels for three biologically relevant genes in a representative sample selected from a population of *A. millepora* under field bleaching conditions. We therefore encourage future studies to consider these genes (*Am*Cat, *Am*Ch and *Am*CTL) for further investigation as key bleaching stress candidates for field-based biomonitoring studies on scleractinian corals.

Conclusion

This study represents the first step towards the application of qPCR technology to monitor and investigate the bleaching response in a widespread Indo-Pacific coral (*A. millepora*) under field conditions. Three candidate ICGs were identified with stable expression levels when the same colonies were sampled under conditions of healthy symbiosis (January 2001) and when in a naturally bleached state (January 2002). Three consistently up-regulated genes of interest (*Am*Cat, *Am*Ch and *Am*CTL) were also identified as key candidate genes for monitoring bleaching stress in *A. millepora*. Future work should focus on evaluating variation in the expression of these GOIs both within individual coral colonies and between *A. millepora* populations.

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